Tan Dat **Nguyen**

PhD Student · Speech and Audio Processing

Daejeon, Republic of Korea

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Research Profile

PhD student at KAIST developing efficient generative models for speech synthesis. Research spans LLM-based TTS, codec language models, diffusion vocoders, and hybrid Mamba–Transformer architectures, with an emphasis on reducing inference latency and memory toward efficient full-duplex spoken dialogue models.

Education

Korea Advanced Institute of Science and Technology (KAIST) — *Daejeon, Republic of Korea*

Ph.D. in Electrical Engineering — *Mar. 2025 - Present*

Advised by Prof. Joon Son Chung, MMAI Lab

Korea Advanced Institute of Science and Technology (KAIST) — *Daejeon, Republic of Korea*

M.S. in Electrical Engineering — *Mar. 2023 - Feb. 2025*

Advised by Prof. Joon Son Chung, MMAI Lab

Ho Chi Minh University of Technology (HCMUT) — *Ho Chi Minh City, Vietnam*

B.S. in Computer Science — *Aug. 2018 - Nov. 2022*

Publications

Underline denotes co-first or co-corresponding authors.

First and Co-first Author Conference Papers

1. **Tan Dat Nguyen**, Sangmin Bae, Joon Son Chung, and Ji-Hoon Kim, "MamTra: A Hybrid Mamba-Transformer Backbone for Speech Synthesis," in *Proc. Interspeech*, 2026. Oral
2. **Tan Dat Nguyen**, Jaehun Kim, Ji-Hoon Kim, Shukjae Choi, Youshin Lim, and Joon Son Chung, "SPADE: Structured Pruning and Adaptive Distillation for Efficient LLM-TTS," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2026. Oral
3. The Hieu Pham, **Tan Dat Nguyen**, Phuong Thanh Tran, Joon Son Chung, and Duc Dung Nguyen, "MAGE: A Coarse-to-Fine Speech Enhancer with Masked Generative Model," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2026.
4. **Tan Dat Nguyen**, Ji-Hoon Kim, Jeongsoo Choi, Shukjae Choi, Jinseok Park, Younglo Lee, and Joon Son Chung, "Accelerating Codec-based Speech Synthesis with Multi-Token Prediction and Speculative Decoding," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2025.
5. **Tan Dat Nguyen**, Ji-Hoon Kim, Youngjoon Jang, Jaehun Kim, and Joon Son Chung, "FreGrad: Lightweight and Fast Frequency-aware Diffusion Vocoder," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2024. Oral
6. **Tan Dat Nguyen**, Quang Tuong Lam, Duc Hao Do, Huu Thuc Cai, Hoang Suong Nguyen, Thanh Hung Vo, and Duc Dung Nguyen, "A Linguistic-based Transfer Learning Approach for Low-resource Bahnar Text-to-Speech," in *Proc. NAFOSTED Conf. Inf. Comput. Sci. (NICS)*, 2022. Oral
7. **Tan Dat Nguyen**, Lam Quang Tuong, and Nguyen Duc Dung, "Calib-StyleSpeech: A Zero-shot Approach in Voice Cloning of High Adaptive Text To Speech System with Imbalanced Dataset," in *Proc. Int. Conf. Applied Business and Data Analytics (ICABDE)*, 2021. Oral

Co-authored Conference Papers

1. Hanna Lee, **Tan Dat Nguyen**, Jaehoon Kang, and Kyuhong Shim, "WAND: Windowed Attention and Knowledge Distillation for Efficient Autoregressive Text-to-Speech Models," in *Proc. Interspeech*, 2026.

2. Jaehun Kim, Ji-Hoon Kim, Yeunju Choi, **Tan Dat Nguyen**, Seongkyu Mun, and Joon Son Chung, "AdaptVC: High Quality Voice Conversion with Adaptive Learning," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2025.
3. Jaemin Jung, Junseok Ahn, Chaeyoung Jung, **Tan Dat Nguyen**, Youngjoon Jang, and Joon Son Chung, "VoiceDiT: Dual-Condition Diffusion Transformer for Environment-Aware Speech Synthesis," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2025.
4. Tuong Q. Lam, Dung D. Nguyen, **Tan Dat Nguyen**, Han K. Lam, Thuc H. Cai, Suong N. Hoang, and Hao D. Do, "Instance-Based Transfer Learning Approach for Vietnamese Speech Synthesis with Very Low Resource," in *Proc. Future of Information and Communication Conference (FICC)*, 2022.

Preprints

1. The Hieu Pham, **Tan Dat Nguyen**, Phuong Thanh Tran, and Duc Dung Nguyen, "Wanna Hear Your Voice: Adaptive, Effective, and Language-Agnostic Approach in Voice Extraction," arXiv preprint, 2023.

Research Experience

Graduate Researcher — *Daejeon, Republic of Korea*

MMAI Lab, Korea Advanced Institute of Science and Technology (KAIST) — *Mar. 2023 - Present*

- Develop efficient speech generative models for real-time synthesis, spanning LLM-based TTS, codec language models, diffusion vocoders, and Mamba–Transformer architectures.
- Reduced codec language model KV-cache use by 66% through multi-token prediction and speculative decoding; investigated structured pruning and knowledge distillation for LLM-based TTS.
- Contributed to research published at ICASSP and Interspeech, including two ICASSP oral presentations.

Research Intern & TTS Researcher — *Ho Chi Minh City, Vietnam*

OLLI Technology Corp — *Jul. 2021 - Feb. 2023*

- Developed Vietnamese text-to-speech systems for voice cloning and low-resource speech synthesis.
- Applied transfer learning and zero-shot voice cloning to build production TTS pipelines with limited training data.

Honors & Awards

2025	**CMK Foundation Global Scholarship**, CMK Foundation	
2023–2025	**Full Scholarship (tuition and stipend) – PhD & Master Courses**, KAIST	*Daejeon, Republic of Korea*
2022	**Certificate of Merit – High Achievements in Academic Competitions & Research**, HCMUT	*Vietnam*
2022	**1st Prize – Poster Competition, Technical Festival**, HCMUT	*Vietnam*
2018	**Vallet Scholarship**, Vallet Foundation	*Vietnam*
2016–2018	**Selected Mathematics Awards – 3rd Prize (VMO and HOMC), Silver and Bronze Medals (Olympiad 30/04)**, National and regional competitions	*Vietnam*

Skills

Programming	Python, C/C++
Frameworks	PyTorch, Hugging Face Transformers, fairseq
Languages	Vietnamese (native), English (proficient; TOEIC 930)